

Lecture 5 Live Problems*The insulin–glucose system. Feedback loops. Parameters.*Points toward lab manual pages: **IG1, IG2, IG3, IG4** (section L1 S4)**1. Glucose, step by step.** Given the change equation

$$\frac{\Delta G}{\Delta t} = m - s I G,$$

suppose that insulin $I = 1$ and sensitivity $s = 0.8$. Suppose that $m = 4$, and $\Delta t = 1$ hour.

- (a) If G is 3 now, what will G be in one hour? In two hours?
- (b) What if $m = 10$? What will be different in the glucose change?
- (c) What if $I = 0$ and $m = 4$ again? What will happen to G in one hour? In two hours?

t (hours)	G	$\Delta G / \Delta t$	ΔG
0	3		
1			
2			